

**MINISTRY OF EDUCATION AND TRAINING
FPT UNIVERSITY**

**Implement A Scenario-Based Web Crawling System
to Serve Decision Support System**

by

Truong Van Long

A thesis submitted in conformity with the requirements
for the degree of Master of Software Engineering

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Supervisor:
Dr. Phan Duy Hung

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Degree Master of Software Engineering

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« Year of Convocation »

Abstract

Along with the strong development of the Internet in recent decades across the globe, the Web data source has become a huge data store for exploiting useful information (according to NetCraft survey in July 2021, there are about 1,216,435,462 websites worldwide [1]). The data source that the Internet brings is becoming more and more massive and diverse, covering all aspects of life from culture, economy, politics, travel, study, research, etc. With such an abundance of resources, the need to search and process information, along with the ability to exploit them effectively, makes it easier and more accurate for people to make informed decisions. Decision-making, business strategy, risk analysis, anti-crisis communication... have become essential needs in modern society. However, the most important element of any information processing is data, without data there would be nothing to analyze and process and no decisions would be made.

Today, companies and businesses increasingly tend to exploit the power of online media. Advertising and marketing campaigns are carried out by applying many different communication channels simultaneously, these campaigns reach very quickly many different user segments, as many people spend most of their time they are online. But communication is also a "double-edged sword" that can help businesses succeed, and can also push businesses down a cliff.

Within the scope of this thesis, I would like to introduce the process of surveying, analyzing, designing, and installing a scenario-based Big Data Collection System. A powerful

system that allows web crawling from various sources (online news sites, forums, e-commerce sites, personal blogs...). The data collection system introduced in this thesis serves as a data source for the Decision Support System (DSS), which helps companies and businesses in reputation monitoring, sentiment analysis, etc. customer contact, thereby offering appropriate communication strategies to prevent and combat communication crisis.

The content of the thesis is organized into five chapters as follows:

Chapter 1: Presents the general theory of decision support systems. This is the basic knowledge about decision support systems such as the decision-making process, the basic components of the decision support system.

Chapter 2: Learn about data crawler concepts, different strategies are being employed in web crawling. Explore and examine some data crawler solution and current challenges.

Chapter 3: Presenting scenario-oriented web-based big data collection solution including system requirements, overall design, main modules.

Chapter 4: Presenting the application of big data collection system in listening to social networks and fighting communication crises. As a concrete demonstration of the practicality of the data collection system.

Chapter 5: Conclusion and Future Work

Acknowledgments

To complete the content of this thesis, I have received a lot of help from my guide, family, agency and colleagues.

First of all, I would like to express my deep gratitude to Mr. Phan Duy Hung - the teacher who directly guided and helped me in the process of building and completing this thesis.

Due to the limitations of time and knowledge, the thesis inevitably has shortcomings. We look forward to receiving valuable comments and suggestions from teachers and fellow students.

Finally, I would like to express my sincere gratitude to my parents, wife, family, and friends who always facilitated, encouraged and helped me enthusiastically to complete the thesis.

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Chapter 1

Decision Support System

The rapid growth of the Internet in recent decades has brought about an enormous amount of data. The amount of data shared on the network is very large with rich and diverse content coming from many different sources such as news websites (providing information related to all aspects of life), in-depth forums about a particular niche, personal blogs, etc. Data sources are rich, but using them effectively to help people, especially those with decision-making roles in organizations and businesses, is a big challenge. The idea of this thesis is to build a scripted web-based big data collection system. The collected data will be used for the Decision Support System, specifically the application of the Social Listening System in fighting communication crises.

1.1 What is Decision Making?

Decision-making on a problem appears in all areas and activities of life without us sometimes realizing it. From simple things like choosing an outfit to go to a party to big tasks like allocating budgets to national programs are decision making.

Thus, decision-making is the process of consciously choosing between two or more alternative courses of action to choose an action that will produce a desired outcome under known constraints [2] [3].

On the basis of the purpose of decision-making activities, the organizational decisions are divided into 3 categories:

- **Strategic Planning Decisions:** Strategic planning decisions are those decisions in which the decision-maker develops objectives and allocates resources to achieve these objectives. Such decisions are taken by strategic planning level (top level) managers.
- **Management Control Decisions:** Management control decisions are taken by management control level (middle level) managers and deal with the use of resources in the organization.

- **Operational Control Decisions:** Operational control decisions deal with the day-to-day problems that affect the operation of the organization. These decisions are taken by the managers at operational level (bottom level) of the organization.

On the basis of the degree of programmability of a decision, Simon proposed two types of decisions [4]:

- **Programmed/Structured Decisions:** Programmed or structured are those decisions, which are well defined and some specified procedure or some decision rule might be applied to reach a decision. Such decisions are routine and repetitive and require little time for developing alternatives in the design phase. Programmed or structured decisions have traditionally been made through habit, by operating procedures or with other accepted tools.
- **Non-programmed /Unstructured Decision:** Contrary to programmed decision making, non-programmed decisions are ill-structured and one-time decisions. Problems or situations that don't have a concrete set of rules or guidelines to follow rely on non-programmed decision making.

1.2 Decision Making Process

Many attempts have been made to describe the human decision-making process. Although each person, based on life experience, offers different streamlined steps for identifying and solving problems and making decisions, common sense suggests that a different approach is needed. general to human decision-making, maybe not in detail, but at a high level of conceptualization. Among the proposed related hypotheses must be mentioned the hypothesis of Newell and Simon (1972) and Simon (1982). Herbert Alexander Simon (1916–2001) was an American economist and political scientist whose primary interest was in human decision-making in organizations. He received the Nobel Prize in Economics in 1978 and the Turing Prize in 1975.

Decision-making is seen as a creative, experiential activity but today it is seen by many as a systematic, evidence-based scientific process. Therefore, in order to improve the likelihood of achieving the best possible outcome, decision-makers should follow a standardized, systematic, and rational decision-making process. Simon points out that the

systematic decision-making process consists of four stages [4]: intelligence, design, choice, and implementation. Simon's model is widely accepted as the shortest, but the most complete model of rational decision making. The figure given below depicts Simon's decision-making model clearly.

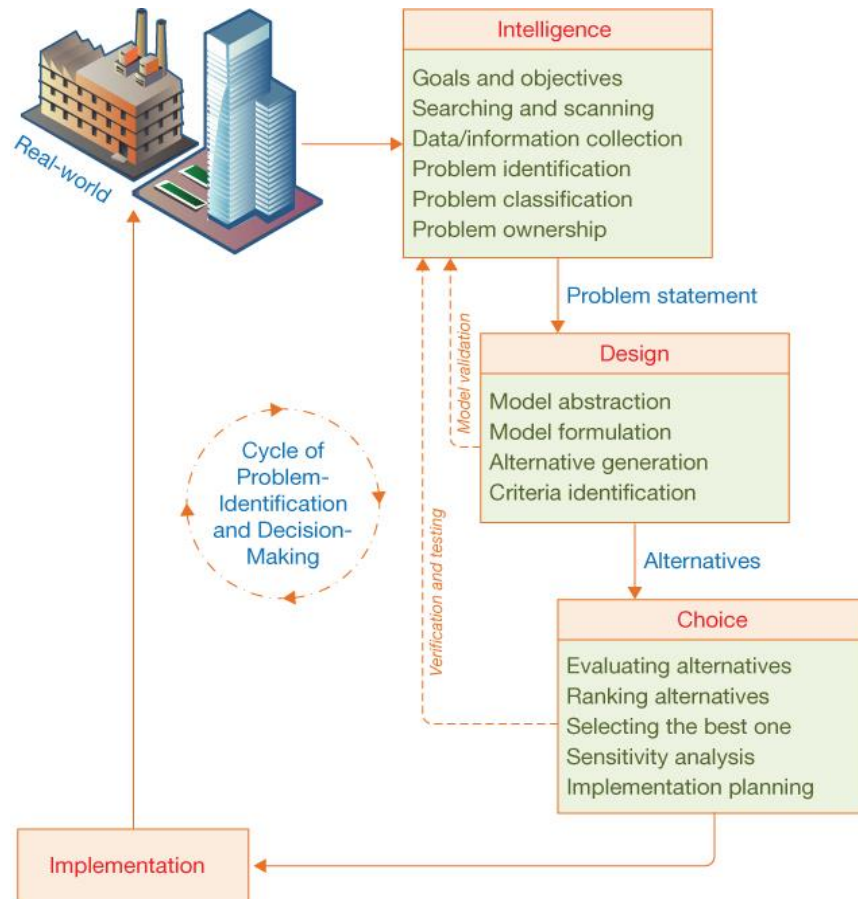


Figure 1 Simon's four step of the decision-making/modeling process.

▪ Phase 1: Intelligence

The decision-making process begins with the discovery phase. In this phase, decision-makers examine the facts and try to pinpoint problems or opportunities. Therefore, the learning phase includes actions such as:

- Define the goals of the organization;
- Search and collect data;
- Identify, identify the subject of the problem, classify the problem statement;

This phase may also include monitoring the results of the implementation phase of a previously completed decision-making process.

- **Phase 2: Design**

The job of this design phase is to base on the problem to be solved and its goals, to propose and analyze different options of action so that a choice can be made based on the available information. Offering different options of action not only serves the purpose of choosing the best option but also avoids bottlenecks in future implementation because alternatives are always available if a decision is made. specific error.

- **Phase 3: Choice**

This is the stage where possible solutions are compared with each other to find the most suitable solution. The “best” solution can be determined using quantitative tools such as decision tree analysis or qualitative tools.

- **Phase 4: Implementation**

Implement the chosen solution, monitor the results, and make adjustments as necessary. In this step, the information system provides operating reports

1.3 Decision Support System

1.3.1 Introduction

In the 1970s, Scott Norton introduced the first concept of Decision Support System (DSS). He defines DSS as “interactive, computer-based systems that help decision-makers use data and models to solve complex, unstructured, fuzzy problems with zero solutions.” complete”. To date, there is no uniform definition of DSS. However, all agree that the most basic purpose of DSS is to assist and improve decision-making.

DSS is a computer-based decision-making tool and it represents a balance between human and computer processing power to solve problems. That's because the nature of each problem is different, some problems are solved more efficiently by computers while humans solve other types of problems more efficiently. Modern computers can solve a problem very

efficiently when the problem is very well structured and when it can be logically formulated using certain rules. On the other hand, unstructured problems are better solved by humans.

1.3.2 Components

The four main components of a DSS framework are:

- **Database Management**

The data management component performs the function of storing and maintaining the information that Decision Support System use. It contains data from various sources, including internal data from the organization, the data generated by different applications, and the external data mined from the Internet, etc. The decision support systems database can be a small database or a standalone system or a huge data warehouse supporting the information needs of an organization. To avoid the interference of decision support system with the working of operational systems, the DSS database usually contains a copy of the production database.

- **Model Management**

Also known as model base management system (MBMS - model base management system) is a software package consisting of components of statistics, finance, management science, or quantitative methods to equip the system. Analytical capabilities, modeling languages. This component can connect to the organization's model repositories or external models.

- **Knowledge Management**

The knowledge management component, like that in an expert system, provides information about the relationship among data that is too complex for a database to represent. It consists of rules that can constrain possible solution as well as alternative solutions and methods for evaluating them.

- **User Interface**

This is a graphical interface that makes it easier for decision makers to interact with the Decision Support System. It displays the results (outputs) of the analysis in various forms,

such as text, tables, charts or graphics. The user can select the appropriate option for viewing on demand.

Decision Support Systems

Components of a Decision Support System

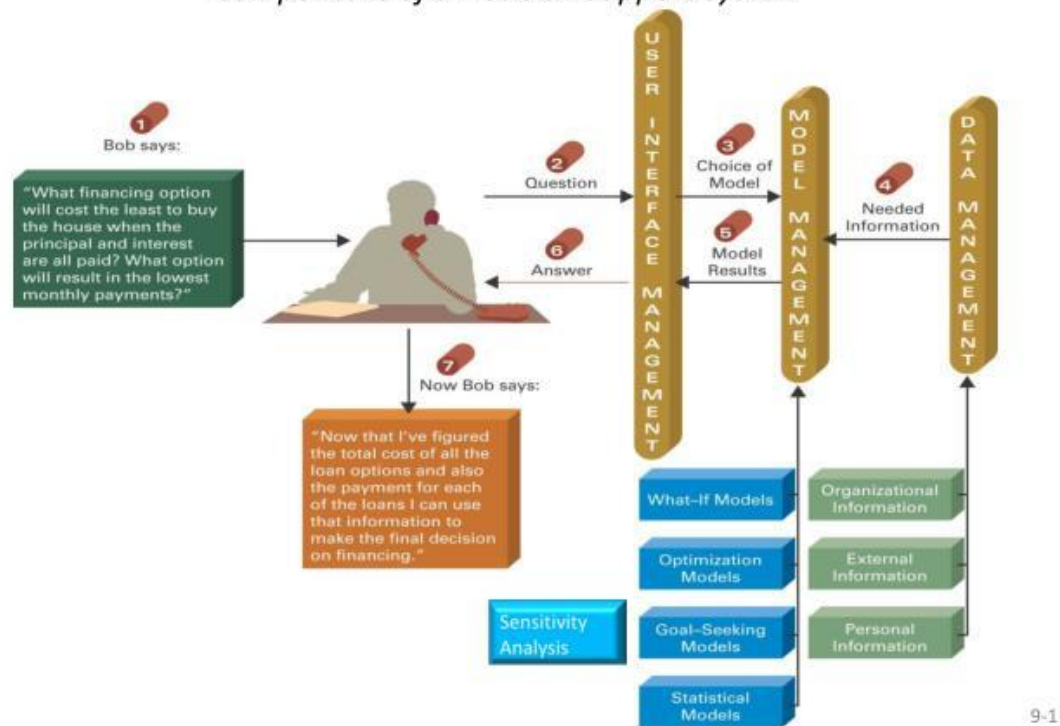


Figure 2 Component of Decision Support System

1.3.3 Decision Making Model

A fundamental feature of DSS is that there must be at least one decision support model. Modeling and selection is part of the second stage (Design Phase) of the decision-making process.

A Quantitative Model usually consists of the following three components:

- **Decision Variables:** These are options determined by the decision-maker. For example, in the investment decision problem, this is the investment amount, investment place, investment time...

- **Uncontrollable Variables:** These are variables that are not in the control of the decision-maker (influenced by external factors). For example, in the above problem, this is the inflation rate, bank interest rate...
- **Result Variables:** These are the outcome variables of the model. For example, in the above problem, this is the profit ratio...

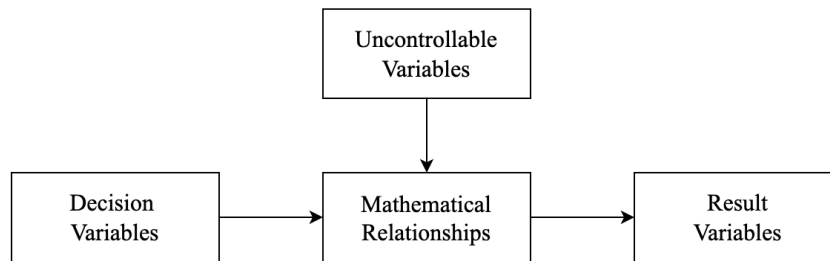


Figure 3 Quantitative Model

When choosing a final decision, the decision-maker may want an optimal decision or a satisfactory, near-optimal decision. Therefore, it can be divided into two types of decision support models:

- **Normative Model:** This model considers all the alternatives and chooses the optimal one.
- **Descriptive Model:** This model considers a set of conditions at the user's discretion and considers a way to manipulate these conditions and produce a satisfactory result. Since this model does not consider all alternatives, the final result may be close to optimal.

Normative models are often used in objective optimization problems. Descriptive models are often used in multi-objective optimization problems when these goals may conflict.

1.3.4 Classification

Decision Support Systems are classified based on many criteria. Currently, there is no uniform classification. These can be categorized into five types:

- **Communication-driven:** Most communications-driven DSSs are targetted at internal teams, including partners. Its purpose are to help conduct a meeting, or for users to collaborate. The most common technology used to deploy the DSS is a web or client

server. Examples: chats and instant messaging softwares, online collaboration and net-meeting systems.

- **Data-driven:** Most data-driven DSSs are targeted at managers, staff and also product/service suppliers. It is used to query a database or data warehouse to seek specific answers for specific purposes. It is deployed via a main frame system, client/server link, or via the web. Examples: computer-based databases that have a query system to check (including the incorporation of data to add value to existing databases).
- **Document-driven:** Document-driven DSSs help managers retrieve and manage unstructured documents and web pages by integrating a variety of storage and processing technologies to provide complete document retrieval and analysis. It also access documents such as company policies and procedures, product specification, catalogs, corporate historical documents, minutes of meetings, important correspondence, corporate records, etc. and are usually driven by a task-specific search engine.
- **Knowledge-driven:** Knowledge-driven DSSs or 'knowledgebase' provide recommendation and/or suggestion schemes which aids the user in selecting an appropriate alternative to a problem at hand. Knowledge driven DSS are often referred to as management expert systems or intelligent decision support systems. They focus on knowledge and recommends actions to managers based on an analysis of a certain knowledge base. Moreover, it has special problem solving expertise and are closely related to data mining i.e. sifting through large amounts of data to produce content relationships.
- **Model-driven:** Model-driven DSSs are complex systems that help analyse decisions or choose between different options. These are used by managers and staff members of a business, or people who interact with the organization, for a number of purposes depending on how the model is set up - scheduling, decision analyses etc. These DSSs can be deployed via software/hardware in stand-alone PCs, client/server systems, or the web.

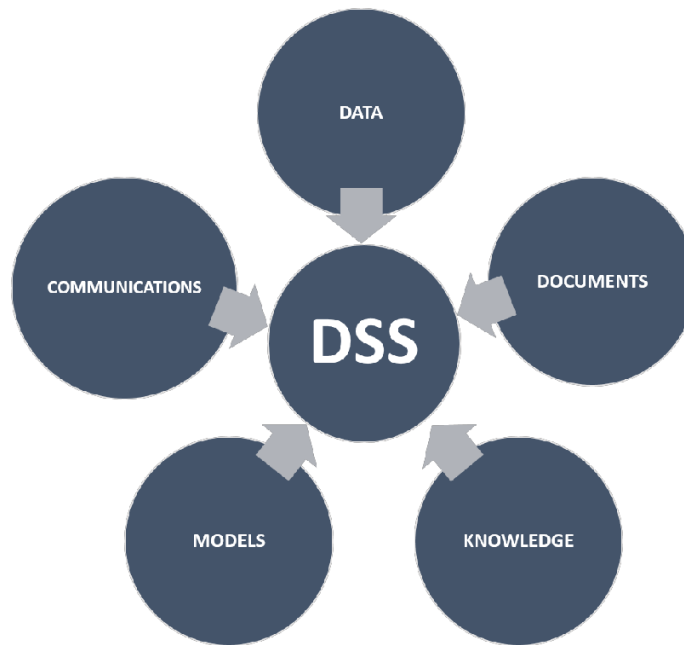


Figure 4 Types of Decision Support System

1.4 Applications

Online auctions are becoming increasingly popular. eBay can be considered the largest online auction house today. The market price of a product is generally nonstationary at eBay — it fluctuates over time. It is even possible that identical items receive different bids at any given point in time. Merchants might buy items at eBay and try to re-sell these items with a profit. The success of these merchants depends on their ability to find bargains and, of course, their bidding strategy. The large profits from the auction of products have attracted researchers interested in this field. A support system for predicting eBay end prices has been introduced in [5]. The end price predictions are based on the item descriptions found in the item listings of eBay, and on some numerical item features.

Within the expansion of the Internet, there has also been a growing interest in online news, which allows an easy and fast spread of information around the globe. Thus, predicting the popularity of online news is becoming a research trend. Popularity is often measured by considering the number of interactions in the Web and social networks (e.g., number of shares, likes, and comments). Predicting such popularity is valuable for authors, content providers, advertisers, and even activists/politicians (e.g., to understand or influence public opinion). The paper [6] proposes a novel and proactive Intelligent Decision Support System (IDSS) that

analyzes articles before their publication. Using a broad set of extracted features (e.g., keywords, digital media content, earlier popularity of news referenced in the article) the IDSS first predicts if an article will become popular.

Predicting football results is a rapidly growing area of academic interest. Economists use models to assess the efficiency of betting markets, operational researchers use models to experiment with the various effects of tournament design, and statisticians showcase their proficiency with advanced statistical techniques by modeling the intricacies of football data. Decision Support System for predicting Football Game results [7] has been introduced aims to support bookmaker's users to increase their profits on bets related to football matches, suggesting to them which bet that they should carry out (home win, draw or away win).

In addition, there are many applications of DSS introduced such as [8] [9]. All of the above applications have one thing in common, they need to prepare data, specifically, collect data on the internet (collect product prices on eBay, collect news, collect football results ...). However, data collection is often performed individually, serving a specific need, lacking an overall solution from start to finish. A solution that anyone from tech-savvy to non-technical people can use. This is also my proposal within the scope of this thesis and will be presented in detail in the following chapters.

Chapter 2

Web Crawler

This chapter presents the general knowledge of Web Crawler. Explore some of the data collection systems covered in scientific articles and study a few products that have been implemented. This is also the basis for building a scenario-based data collection system.

1.5 Definition

A web crawler (also known as a crawling agent, a spider bot, web crawling software, website spider, or a search engine bot) is a program that goes around the internet collecting and storing data in a database for further analysis and arrangement [10]. The process of web crawling involves gathering pages from the web and arranging them in such a way that the search engine can retrieve them efficiently. The critical objective is to do so efficiently and quickly without much interference with the functioning of the remote server. A web crawler begins with a URL or a list of URLs, called seed URLs [11]. The crawler visits the URL at the top of the list. On the web page it looks for hyperlinks to other web pages, it adds them to the existing list of URLs in the list. This methodology of the crawler visiting URLs depends on the rules set for the crawler. In general crawlers incrementally crawl URLs in the list. In addition to collecting URLs the main function of the crawler, is to collect data from the page. The data collected is sent back to the home server for storage and further analysis.

Crawling can be viewed as a graph traversal problem. The entire web world is viewed as a large graph with nodes as web pages and links as paths (edges). A crawler starts at a few kernel nodes and then follows the edges to reach other nodes. The process of loading a web page and extracting the links in it is similar to expanding a node in a graph search problem. An intentional crawler tries to follow edges that are expected to lead to locations in the graph that are valid for that topic.

1.6 Architecture

The overview architecture of web crawlers can be summarized as shown in Fig.5.

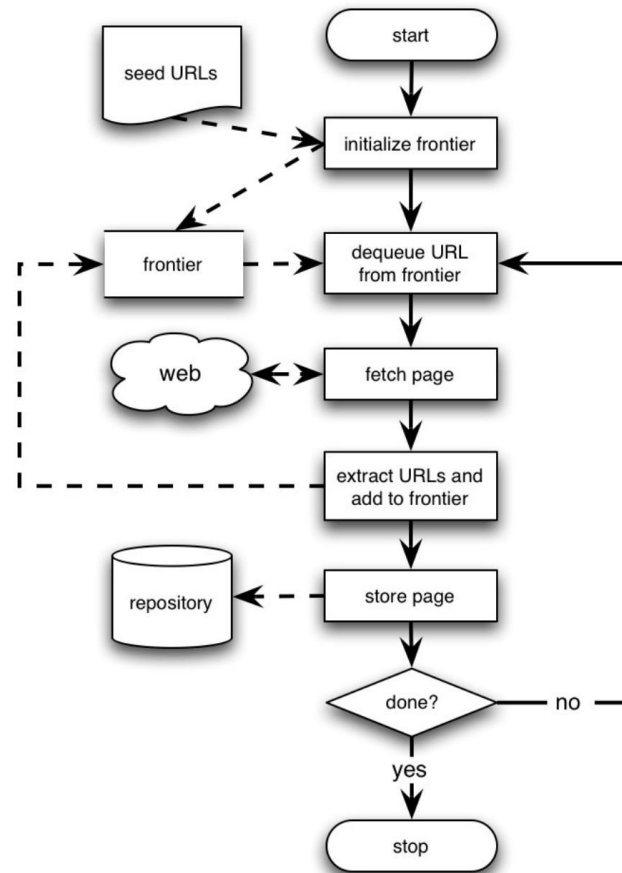


Figure 5 An overview of web crawler architecture

Figure 5 shows the generalized architecture of web crawler. It has three main components [12], [13]: A **frontier** which stores the list of URL's to visit, **Page Downloader** (fetch page) which download pages from WWW and **Web Repository** (store page) receives web pages from a crawler and stores it in the database. Here the basic processes are briefly outline:

- **Crawler frontier** - It contains the list of unvisited URLs. The list is set with seed URLs which may be delivered by a user or another program. Simply it's just the collection of URLs. The working of the crawler starts with the seed URL. The crawler retrieves a URL from the frontier which contains the list of unvisited URLs. The page corresponding to the URL is fetched from the Web, and the unvisited URLs from the page are added to the frontier. The cycle of fetching and extracting the URL continues until the frontier is empty

or some other condition causes it to stop. The extracting of URLs from the frontier based on some prioritization scheme.

- **Page downloader** - The main work of the page downloader is to download the page from the internet corresponding to the URLs which is retrieved from the crawler frontier. For that, the page downloader requires a HTTP client for sending the HTTP request and to read the response. There should be timeout period needs to set by the client in order to ensure that it will not take unnecessary time to read large files or wait for response from slow server.
- **Web Repository** - It use to stores and manages a large pool of data "objects," in case of crawler the object is web pages. The repository stores only standard HTML pages. All other media and document types are ignored by the crawler. It is theoretically not that different from other systems that store data objects, such as file systems, database management systems, or information retrieval systems. However, a web repository does not need to provide a lot of the functionality like other systems, such as transactions, or a general directory naming structure. It stores the crawled pages as distinct files. And the storage manager stores the up-to-date version of every page retrieved by the crawler

1.7 Types of Web Crawler

Web crawlers are updated continuously, different strategies are applied to the development of the web crawler. We reviewed the previously published studies and summarized the common characteristics of existing web crawlers [11], [12], [14] . Five types of web crawlers have been summarized as follows.

- **Traditional Crawler** - A traditional crawler periodically crawls the already crawled URLs and replaces the old documents with the newly downloaded documents to refresh its collection. [13]
- **Focused Crawler** - Focused Crawler is also known as Topical Crawler. It collects the documents that are focused and relevant to a given topic. The goal of the focused crawler is to selectively seek out pages that are relevant to a pre-defined set of topics. The topics are specified not using keywords, but using exemplary documents. A focused crawler

analyzes its crawl boundary to find the links that are likely to be most relevant for the crawl, and avoids irrelevant regions of the Web.

- **Incremental Crawler** - The incremental web crawler is a crawler that incrementally updates downloaded web pages and crawls newly generated or changed web pages. Incremental web crawlers do not re-download pages that have not changed. It can effectively reduce data downloads, update crawled web pages in time, and reduce time and space consumption.
- **Distributed Crawler** - Distributed web crawling is a distributed computing technique. Many crawlers are working to distribute in the process of web crawling, in order to have the most coverage of the web. A central server manages the communication and synchronization of the nodes, as it is geographically distributed. The benefit of distributed web crawler is that it is robust against system crashes and other events, and can be adapted to various crawling applications.
- **Parallel Crawler** - Parallel crawlers are multiple crawlers running at the same time. It consists of multiple crawling processes called as C-procs which can run on network of workstations. The Parallel crawlers depend on page selection and page freshness. A Parallel crawler can be distributed at geographically distant locations or be on a local network. Parallelization of crawling system is vital from the point of view of downloading documents in a reasonable amount of time.

1.8 Challenges of Web Crawler

▪ **Challenge 1** Non-Uniform Structures

The web is a dynamic space with inconsistencies in data formats and structures. There are no norms to be followed while building a web presence. Due to this lack of uniformity, collecting data in a machine-readable format can be difficult. The problem gets amplified with scale when you need structured data. This places challenges when a lot of details are to be extracted pertaining to a specific schema from thousands of web sources.

▪ **Challenge 2:** Omnipresence of AJAX Elements

AJAX and interactive web components make websites more user-friendly. But not for crawlers. The webpage content is produced dynamically (and on-the-go) by the browser and therefore not visible to crawlers. Crawlers need to be refined in their approach to be more efficient and scalable.

- **Challenge 3:** Bandwidth and Impact on Web Servers

One of the biggest challenges or limitations faced by web crawlers is the high consumption rate of network bandwidth. This can particularly happen when the web crawler downloads many irrelevant web pages. To maintain the freshness of the database, crawlers adopt a polling method or use multiple crawlers, which consumes a lot of bandwidth. If a web crawler is frequently visiting websites, then the performance of the web servers will be severely impacted.

- **Challenge 4:** The Rise of Anti-Scraping Tools

Today, web developers have tools such as ScrapeSheild and ScrapeSentry that can differentiate bots from humans. Using such tools, web developers can manipulate content shown to bots and humans, and also restrict bots from scraping the website.

1.9 Literature Review Results

There are many crawlers written in every programming and scripting language to serve a variety of purposes depending on the requirement, purpose and functionality for which the crawler is built. All crawlers are trying to solve one of the problems mentioned in section 1.8. Here are a few solutions that have been published recently.

Nowadays, webpages are non-uniform structures, that why most extraction methods cannot adapt to various and heterogeneous web structures and have poor generality and extracting efficiency. Xin Yu & Zhengping Jin has proposed a method that can adapt to the heterogeneity and variability of web pages and gets high precision and recall [\[15\]](#). Their method is based on DOM structure to divide one web page into several blocks, and extract content blocks with statistical information instead of machine learning repeating training and manual labeling, which gets good performance in Precision, Recall, and F1.

Also, the content of the website is dynamic, dependent on user interaction. It is the use of technologies for dynamically loading web content such as Asynchronous JavaScript and XML (AJAX). Solving this problem is the subject of much of current research into crawling systems and has led to the development of the dynamic web crawler, also known as event-driven web crawler. AJAX-based web applications and Single Page Application (SPA) [16] architectures are increasingly gaining popularity for developing dynamic websites. These new architectures provide more sophisticated client-side processing of the web document the user interacts with compared to the traditional server-side preprocessed web document [17].

Due to the huge amount of data that crawlers need to process, distributed crawlers are also an active area of research to improve the performance of crawler systems [18], [19]. Distributed crawlers improve the performance of a normal crawler system by using a distributed architecture. Distributed crawlers may use one or several other crawler classes.

Most current solutions focus on dealing with technical problems, with little concern for end-users. They are people who have a need to collect information on the web but have little technology knowledge (they can be salespeople, students, analysts ...). From that need, I propose an end-to-end crawler solution called "Scenario-Based Web Crawler System". The system must be able to:

- **User-oriented:** People with little technical knowledge can still perform data collection easily on a desktop application.
- **Connection-oriented:** Aim to eliminate data collection work on data mining/analysis systems. The system is designed so that it is easy to integrate with other systems
- **Solve technical problems:** Dynamic content, deep web, ajax, login, huge amount of data...

Specific design and implementation of the solution will be described in the following chapter.

Chapter 3

Scenario-based Web Crawler System

Web scraping is the process of extracting information from the world's largest database: the world wide web. Data gathering projects that would otherwise require hundreds of work-hours can be easily and quickly done using a web scraper. As Big Data and Data Science become increasingly important, more organizations are realizing the potential of web scraping as vital to data-driven decision making. Common web scraping use cases include:

- Product listing
- Market research
- Competitor research
- Brand health assessment
- Social listening

A good web scraping tool can collect almost any data point on the internet, so potential use cases are almost endless – no matter what industry you work in, web scraping can become the ultimate secret weapon for your business or organization.

This chapter presents a Scenario-based Web Crawler system that allows non-technical user to collect data from any webpage just by some click.

1.10 System Requirements

The Scenario-based Web Crawler System is a general-purpose tool for crawling and extracting arbitrary semi-structured data from social sources such as news, community, and blogs and transforms them into a structured format for later analysis and retrieval. For example, users can extract product information, product reviews from online shopping malls, or, users can extract discussion messages from a social network, or a bulletin board. This system offers the best big data collection environment for real-time big data analysis, competitor analysis, market and product analysis, risk management, and customer voice analysis, includes four highlights as follows:

- **Data Management:** The system enables the user structure, organize and prepare the data files for later publishing. The user can export the files directly into, CSV, XML, or JSON and has the option to filter the data.
- **Data identification and downloading:** The system helps user extract text, URLs, images, files from various web pages and transforms them into a structured format.
- **Identify Pages:** Almost all websites implement a listing approach to organize and structure the contents. The system can identify these lists and creates agents that gather the information you specify across various pages. Complex paging structures and nested categories are challenging to work with.
- **Cleaning text and HTML:** The system enables the user to get articles, product descriptions, discussion threads, and image captions in pure texts and sanitized HTMLs.
- **Structured Search:** The user can search for content that is structured from any crawl and return only the results that are matching. All crawls can be searched instantly and allow the user to slice and dice their data by examining the structured fields. The user can sort data by date of the article, filter products by price, and search across different custom fields.

Scenario for extracting Web data

Knowledge extraction requires fine-grained metadata extraction from dynamically auto-generated Html pages such as title, price, date, etc. Which metadata to be extracted must be defined by the user and in order to maintain low costs, any user even without strong technical skills should be able to define rules just knowing how to use a Web browser. The system can capture any actions of the user such form filling, clicking on the submit button, extracting the elements highlighted by the user, etc. through a simple WYSIWYG interface and with few clicks, the user can generate rules for extracting exactly any information he/she wants from any dynamic web page. For example, the user can get search results from a search engine into a tabular form, or extract a list of products, blog posts, or all the discussion messages of a topic from a community, etc.

A scenario is defined as a sequences of navigation actions defined by the user that can be expressed in form of script like: LoadPage(), SetElementValue(), ClickElement()... which are actions to control a Web Browser responsible for capturing dynamic HTML, and also a set of extraction actions like: GetElementValue(), GetListOfElementValue()... which are actions to control a Wrapper responsible for extracting desired metadata. The system should be able to generate rules for extracting data from web pages including texts, URL, images and binary files.

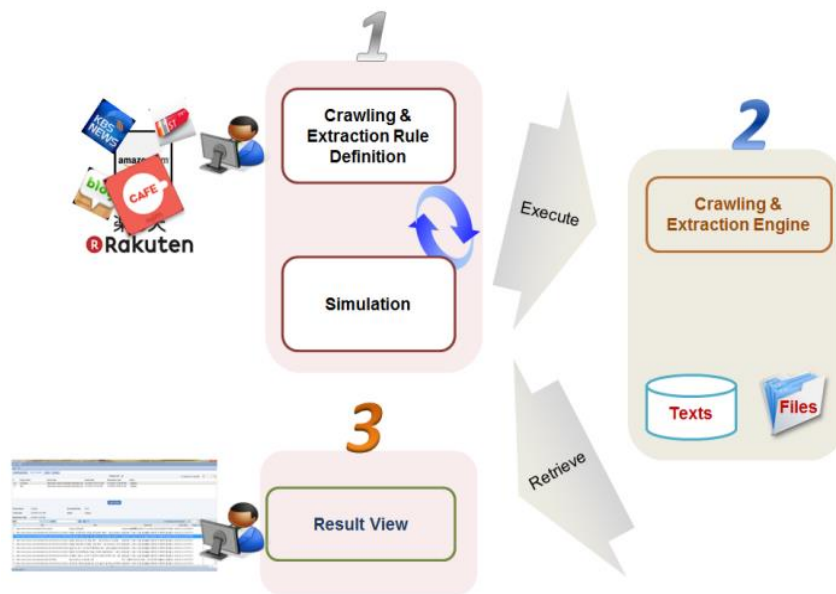


Figure 6 Concept Diagram of Scenario-based Web Extractor

Figure 5 illustrates the scenario-based web crawling, consisting of 3 main steps as follows:

- 1) The user defines a rule and can preview its result through the simulator.
- 2) The user then executes the crawling and extraction engine to extract and store the data. The data is can be text, images, pdf files, audio files, etc.
- 3) The user then can retrieve the crawled data via Workbench or Web Application.

1.11 System Design

1.11.1 Layered architecture

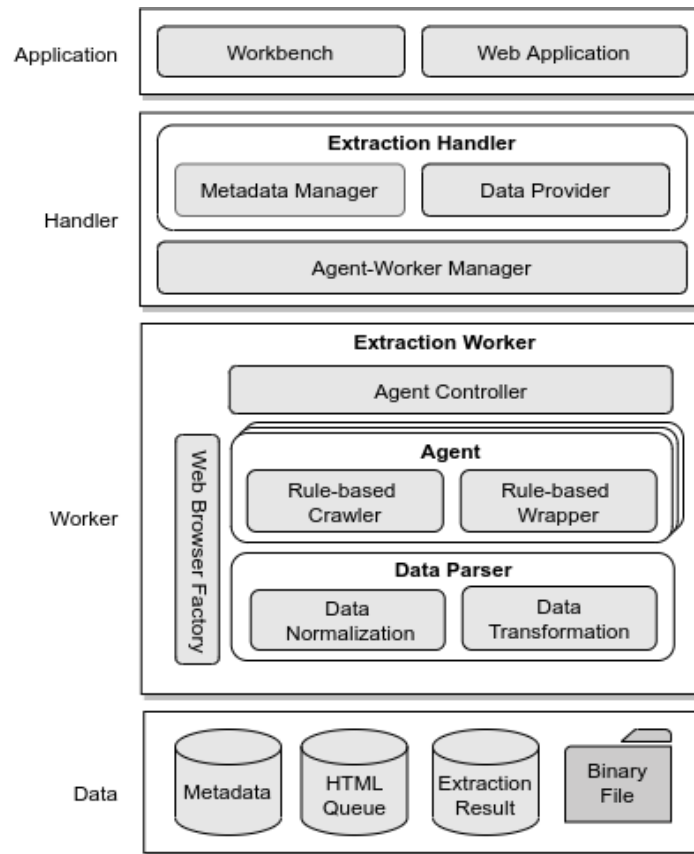


Figure 7 Layered Architecture

Figure 7 is the layered architecture of the crawling and extraction system. The architecture includes the horizontal four layers (Application, Handlers, Workers, Data). Each blue rounded rectangle is a separate component that can run a single process. The system is designed for horizontal scaling, each component can be deployed on separate computers and use a load balancer in case of necessity.

- **The top layer** is the Application layer where various types of applications can be implemented. Workbench is a program that is used to define new scenario-based web data extraction task. It supports the following actions: creating, modifying, and deleting a scenario-based web data extractor task. A workbench provides an intuitive Graphical User Interface enabling users to define easily a new extracting task. It also provides a simulator for testing the scenario-based web extractor task before executing it. The user can view the crawling and extraction result through both Workbench and Web Application.

- **The 2nd Layer** is the Handler layer consisting of two main components: Extraction Handler and Agent-Worker Manager. Extraction Handler processes all requests related to scenario-based data web extraction for extracting various types of data from various social data sources; The handler manages its metadata through its Metadata Manager sub-component. Metadata includes information about the scenario-based web extractor and web crawler tasks (task Id, status, createDate, configuration/rule, etc.). Data Provider in each handler processes requests for retrieving, searching, and exporting crawling and extraction results. Each Handler controls several Workers where the real scenario-based and general web crawler tasks are executed. The Agent-Worker Manager is to manage the mapping between scenario-based web data extraction tasks and the Worker. In other words, Agent-Worker Manager plays a role as a load balancer for the Workers. The handler uses information from Agent-Worker Manager to decide where a new agent (task) will be allocated to.
- **The 3rd Layer** is the worker's layer that controls and executes scenario-based web data extraction. Agent Controller in the Extraction Worker is to control the tasks. The agent sub-component in the Extraction Worker is to execute the scenario-based web extraction task. Each Agent sub-component has rule-based Crawler and rule-based Wrapper for crawling and extracting data through rules. The rule-based Crawler is a browser-based crawler. Thus, the Web Browser Factory provides a browser for the rule-based Crawler.

1.11.2 Scenario-based Web Extractor

The scenario-based web data extraction consists of crawlers and XPath-wrapper. The crawler and wrapper must reproduce the same scenario as the user scenario. The Web Data Extraction system has 2 main components: one is a crawler which is in a headless browser (a web browser without a graphical user interface) that can simulate all the user actions during web page navigation including mouse event and keyword event and the other is an XPath-Wrapper that extracts the information wanted. The XML Path Language (or, briefly, XPath) provides a powerful syntax to address specific elements of an XML document (and, to the same extent, of HTML Web pages) in a simple manner. XPath has been defined by the World Wide Web Consortium, so as DOM [20]

The crawler assures that the correct HTML is given to the XPath-wrapper for extracting the information. With XPath [21], we can identify a single element in the document tree, or address multiple occurrences of the same element. Because a node in XML tree corresponds always ambiguously to several possible XPaths, we propose an approach to select the best XPath among them using annotation information from the user.

Scenario	Rule-based Crawler	HTML Queue	Rule-based Wrapper	Extraction Result
1. Open URL	Control browser to open URL	Empty		Empty
2. Enter Keyword Search	Control browser to enter keyword search	Empty		Empty
3. Click on Search button & Search Result is showed	Control browser click on Search Button Search result is loaded in browser	Empty		Empty
4. Click on 1 st item	Control browser to click on 1 st item page	Empty		Empty
4.1. Item page is opened	The 1 st item page is loaded in browser Add page HTML to HTML Queue	HTML of the 1 st item page		Empty
4.2. Extract data in item page		Empty	Get HTML of the 1 st item page Extract data	Extracted data of the 1 st item page
5. Click on 2 nd item	Control browser to click on 2 nd item page	Empty		Extracted data of the 1 st item page
5.1. Item page is opened	The 2 nd item page is loaded in browser Add page HTML to HTML Queue	HTML of the 2 nd item page		Extracted data of the 1 st item page
5.2. Extract data in item page		Empty	Get HTML of the 1 st item page Extract data	Extracted data of the 1 st and 2 nd item page

Figure 8 Concept of Crawler and Wrapper

Figure 8 describes the collaboration of both crawler and XPath-wrapper when the user scenario is defined. The crawler and wrapper work in pipeline mode. The crawler keeps crawling data and storing raw data (HTML page) to storage called HTML Queue. The Wrapper keeps extracting data from the HTML Queue. Based on each step of the user-scenario, either the Crawler or the Wrapper are involved. For example, in Step 1, the user opens a URL, the Crawler also opens the URL to reproduce a user's activity. In Step 2, the user enters a search keyword, the Crawler also fills the search keyword to the search box. In Step 4, the user clicks on the first item, the Crawler also opens the detail page of the first item. In Step 4.2, the user extracts data in detail page, the Wrapper extracts data from the HTML page of the Step 4.1 through HTML Queue.

Scenario	Extraction Rule Definition
1. Open URL	<pre> <Action> <ActionType>LoadPage</ActionType> <PageUrl>http://kin.naver.com/index.nhn</PageUrl> </Action> </pre>
2. Enter Keyword Search	<pre> <Action> <ActionType>SetElementValue</ActionType> <Value>북한</Value> <ItemXPath>//input[@id="nx_query"]</ItemXPath> </Action> </pre>
3. Click on Search button & Search Result is showed	<pre> <Action> <ActionType>ClickElement</ActionType> <Page>1</Page> <ItemXPath>//form[@id="topSearch"]/fieldset/div[1]/input</ItemXPath> </Action> </pre>
4. Click on 1 st item	<pre> <Action> <ActionType>ClickElement</ActionType> <Page>1</Page> <DestinationPage>2</DestinationPage> <ItemXPath>dl[1]/dt[1]/a[1]</ItemXPath> </Action> </pre>
4.1. Item page is opened	<pre> <Action> <ActionType>PagePlaceholder</ActionType> <Page>2</Page> <LoadUrl>http://kin.naver.com/qna/detail.nhn?d1id=6&dirId=60602</LoadUrl> <IsFrame>False</IsFrame> </Action> </pre>
4.2. Extract data in item page	<pre> <Action> <ActionType>GetElementValue</ActionType> <FieldExpression>Question</FieldExpression> <ItemXPath>//div[@id="contents_layer_0"]/div[1]</ItemXPath> </Action> </pre>

Figure 9 Sample Rule for Rule-based Crawler and Wrapper

The rule should capture all user activities while the user surfs the internet including (1) keyboard event (for example: such as user enter search keywords, user name, password), (2) mouse event (for example the user clicks on a button/link). The rule should be able to define extraction activity for extracting URL, for extracting text, for extracting a list of records (post title, post data, post content), for extracting images, and for extracting screenshot. There are also many cases where web pages display only a part of the data, the extraction rule should be able to define page navigation for exploring more data. Currently, many web pages use JavaScript/frames that prevent the general web crawler from downloading data directly. The rule should be able to work with this case. For readability, the rule should be in JSON format or XML format. Figure 9 is rule sample for the below scenario.

1.11.3 Identify elements

One of the most exploited features in Web Data Extraction is the semi-structured nature of Web pages. The representation of a Web page by using a labeled ordered rooted tree is usually referred to as DOM (Document Object Model). The document tree (henceforth also referred to as DOM tree) has been successfully exploited for Web Data Extraction purposes by a number of techniques in the following.

One of the main advantages of the adoption of the Document Object Model for Web Content Extraction is the possibility of exploiting some tools typical of XML languages (and

HTML is to all effects a dialect of the XML). In particular, the XML Path Language (or, briefly, XPath) provides with a powerful syntax to address specific elements of an XML document (and, to the same extent, of HTML Web pages) in a simple manner. XPath has been defined by the World Wide Web Consortium, so as DOM.

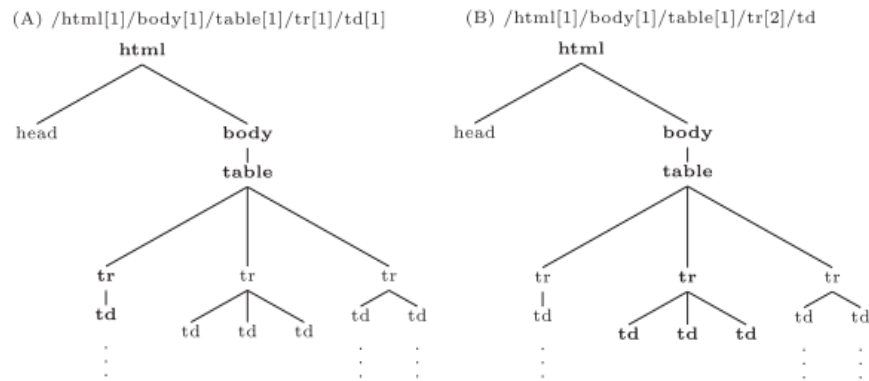


Figure 10 Example of XPath(s) on the document tree

▪ Addressing a element in the document tree

We want to generate XPath from node name table to the first node name tr. There are different ways to generate XPath, Table 1 lists four possible ways. Each way has its own importance. We define α as the importance of the XPath from a node to its child. In general case, there are multiple nodes on the way from an ancestor node to its descendant node. Suppose that we have to find an XPath from A to An. There exists many XPath from A to An.

Algorithm Xpath generation for one element

Input: Node T in DOM document

Output: Best Xpath to T

Algorithm:

Step 1: Look for ancestor A of T that A has ID for A is root node

Step 2: Generate all possible Xpath from A to T

Step 3: Select best Xpath

Figure 11 Algorithm for generating XPath for one element

▪ Addressing the list of elements in the document tree

In order to capture a list of items, the user have to select two items, then, we will automatically identify the boundary of this item.



Figure 12 Capture a list of items

Figure 12 is an example we want to capture the articles from search result by “나의 iPhone 찾기” keyword. There are two steps. Step 1: User select the first article title. Step 2: User select second article title. We will auto understand the list of article user wanna collect by use data wrapper algorithm.

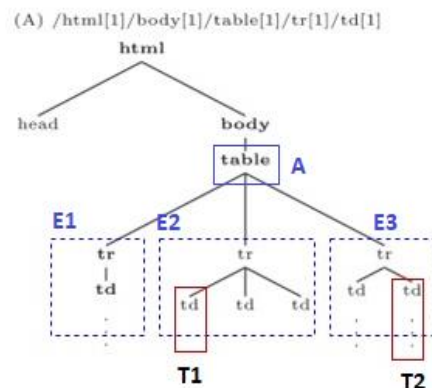


Figure 13 Example of snippet HTML

For simple case, assuming that the user selects two `<td>` element in Figure 13, the `<tr>` nodes in blue boxes are boundary item. Follow Figure 14 for more details on algorithm.

Algorithm Xpath Generation For List of Elements

Input: T1 and T2 are two nodes in a list of an XML document

Output: XPath to extracts a list of nodes containing T1 and T2

Algorithm:

- Step 1. Look for ancestor A of T₁ and T₂ so that A has ID for A is the root node
- Step 2. Generate XPath from root to A using algorithm defined in Figure 11
- Step 3. E1, E2 that are ancestors of T1, T2, respectively.
- Step 4. Select the XPath that has the highest weight as the best XPath

Figure 14 Algorithm XPath Generation for list of elements

1.11.4 Data Transformation

Data from tables can be complicated but system's technology allows it to scrape all the significant information from table structures rather easily. For most of data extraction tools, retrieving data from tables is nearly impossible. What usually takes days is mere minutes.

	PT 909, PT 911	PT 940	PT 945
Type	Double Action semiautomatic		
Caliber(s)	9×19 Luger / Parabellum	.40 S&W	.45 ACP
Weight unloaded	817 g	820 g	855 g
Length	184 mm	178 mm	190 mm
Barrel length	102 mm	92 mm	108 mm
Magazine capacity	15 or 17 rounds	10 or 12 rounds	8 rounds

Figure 15 Table data from website

```

"Models" : [
  {
    "Model" : "PT 909, PT 911",
    "Characteristics" : [
      {
        "key" : "Type",
        "value" : "Double Action semiautomatic"
      },
      {
        "key" : "Caliber(s)",
        "value" : "9×19 Luger / Parabellum"
      },
      ...
    ]
  },
  {
    "Model" : "PT 940",
    "Characteristics" : [
      {
        "key" : "Type",
        "value" : "Double Action semiautomatic"
      },
      {
        "key" : "Caliber(s)",
        "value" : ".40 S&W"
      },
      ...
    ]
  }
]

```

Figure 16 The table data after transforms to structured format

1.11.5 Data Normalization

When scraping dates from the web, we need them in a structured format so we can easily search, sort, and compare them. Dates written in natural language are not suitable for this. For example, the "24th of December" shows up first if you sort the "25th of November" and the "24th of December" alphanumerically. We solve this by taking the natural language date and parsing it into a DateTime object. A bonus perk of Date parser is that you don't need to worry about translation. It supports a range of languages including English, Korean, Japan, and Chinese. Better yet, Date parser auto detects languages, so you don't need to write any additional code.

There are two main approaches to detecting any human-readable representation of a discrete quantity within the text:

- Detect well known and stable patterns in the input stream and by adjacency determine the output stream.
- Windowing through the text in the input stream and directly detect the quantities.

Our approach is hybrids of these. It is a deterministic rule-based system designed for extensibility built on regular expression patterns. We define some types of Expressions in order to recognize the date component follow the abstract syntax tree.

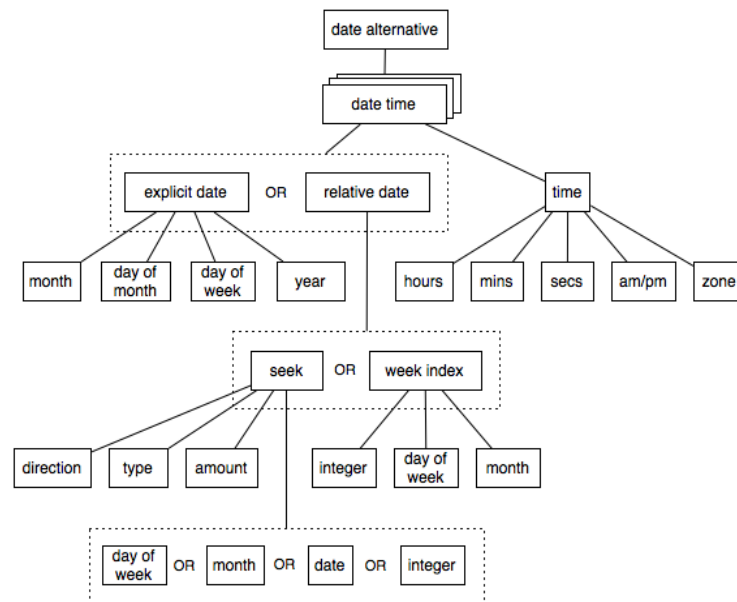


Figure 17 The abstract datetim syntax tree

There are three types of expressions:

- **Duration:** The amount of intervening time between the two end-points of a time interval. Durations can be specified as a combination of a unit (e.g., day, month, year, etc.) and a numeric value (the quantity associated with the unit). Examples of temporal expressions corresponding to the three different types of durations are given below:

Expression	Pattern
5 days ago	{ "pattern": "\\d+\\day[s]?s+Ago", "unit": "DATE" }
6 hours ago	{ "pattern": "\\d+\\hour[s]? \\s+Ago", "unit": "HOUR" }
6시간전	{ "pattern": "\\d+\\s*시간\\s*전", "unit": "HOUR" }
40秒钟前	{ "pattern": "\\d+\\s*秒钟前", "unit": "SECOND" }

Table 1 Pattern for duration time

- **Formal Dates:** A time point indicating a particular instance on a time scale. Recognizes absolute times in standard formats, such as *Dec 31, 2016, 8:20 AM* or *05 Nov at 09:27:58 EDT*. Below, we give examples of some expressions that it can recognize.

Expression	Pattern
Dec 31, 2016, 8:20 AM	{ "pattern": "\\^[a-zA-Z]{3}\\s+[0-9]{1,2},\\s+[0-9]{4},\\s+[0-9]{1,2}:\\s+[0-9]{1,2}\\s+(am pm)\$", "format": "MMM dd,' yyyy',' HH:mm a" }
Oct 27, 17 at 03:01 PM	{ "pattern": "\\^[a-zA-Z]{3}\\s+[0-9]{1,2},\\s+[0-9]{2}\\s+at\\s+[0-9]{1,2}:\\s+[0-9]{1,2}\\s+(am pm)\$", "format": "MMM dd,' yy 'AT' HH:mm a" }
2017. 11. 28.	{ "pattern": "\\^[0-9]{4}\\s+(0[1-9] 1[0-2])\\s+\\s+[0-9]{1,2}\\s+\\s+\$", "format": "yyyy.' MM.' dd.'" }
05 Nov at 09:27:58 EDT	{ "pattern": "\\^[0-9]{1,2} [a-zA-Z]+ at [0-9]{1,2}:\\s+[0-9]{1,2}:\\s+[0-9]{1,2} [a-zA-Z]+\$", "format": "dd MMM 'at' HH:mm:ss z" }

Table 2 Formal Date Pattern

- **Relaxed Date:** Relaxed dates are those in which the month, day of weeks, day of month, and year may be given in a loose, non-standard manner, with most parts being optional, such as “*Post at last Friday at 9 am*”, “*yesterday (19:07)*” or DateTime string contain unwanted content like “*26 March 2018 |*” or “*| 26 March 2018*”.

Expression	Pattern
Last Friday at 9am post at last friday at 9 am	<pre>{ "regexs": [["(?:({NAMED_RELATIVE_DATE})\\s+)(?DAY_OF_WEEK)(?:\\.\\s+)([0-9]{1,2})(?:\\s*)({AM} {PM})?(?\\W \$)"], "groups": { "1": "NAMED_RELATIVE_DATE", "2": "DAY_OF_WEEK", "3": "HOUR", "4": "AMPM" }, "unit": "WEEK", "locale": "en" }</pre>
yesterday 17:00 yesterday at 6:00 pm today 5:30 tomorrow 3 pm Yesterday (19:07)	<pre>{ "regexs": ["({NAMED_RELATIVE_DATE})(?:\\s*.*)\\s+([0-9]{1,2}):([0-9]{1,2})(?:\\s*)({AM} {PM})?(?\\W \$)", "({NAMED_RELATIVE_DATE})(?:\\s*.*)\\s\\S+([0-9]{1,2}):([0-9]{1,2})"], "groups": { "1": "NAMED_RELATIVE_DATE", "2": "HOUR", "3": "MINUTE", "4": "AMPM" }, "unit": "DAY", "locale": "en" }</pre>

Table 3 Relaxed Date Pattern

1.11.6 Multiple browsers supported

Different browsers read the website code differently. That is Browser Compatibility. The term browser compatibility refers to the ability of a certain website to appear fully functional on different browsers that are available in the market. This means that the website's HTML coding, as well as the scripts on that website, should be compatible to run on the browsers. A website's compatibility plays a huge role in the success of scraping. Because, if the browser fails to read the HTML coding of the website, either the website does not appear as it is supposed to on that browser. This adversely affects the quality of data collection. To overcome that, the my system supports many different types of browsers during data collection with a few simple configurations.

- Headless browsers including PhantomJS, Chrome headless and HtmlUnit.
- GUI browsers including IE & Firefox
- HTTP Client

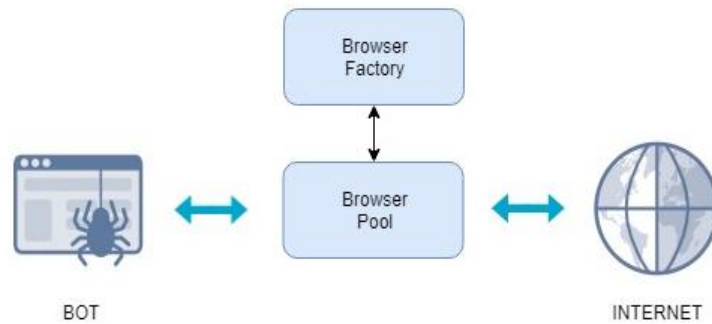


Figure 18 Browser Pool

1.12 Scenario Builder example

Consider the case that a person has a need to buy a used car, they need to survey prices on online car markets. Assume that the desired user scenario is as follows:

- Open url <https://choxeotofun.net/>
- Browse information for each vehicle in the list, extract vehicle name information, short description, vehicle price



Figure 19 User scenario - Capture vehicle information

- Open view information of each post, extract seller information, vehicle details.
- Browse the next pages till the end of the list

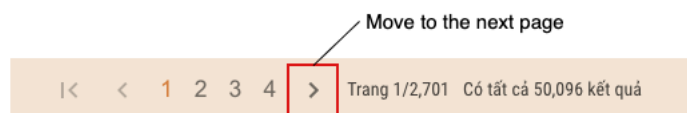


Figure 20 User scenario - Move to next page

As depicted in Figure 6, the user data collection process will go through the following three steps.

✓ **Step 1: Scenario definition**

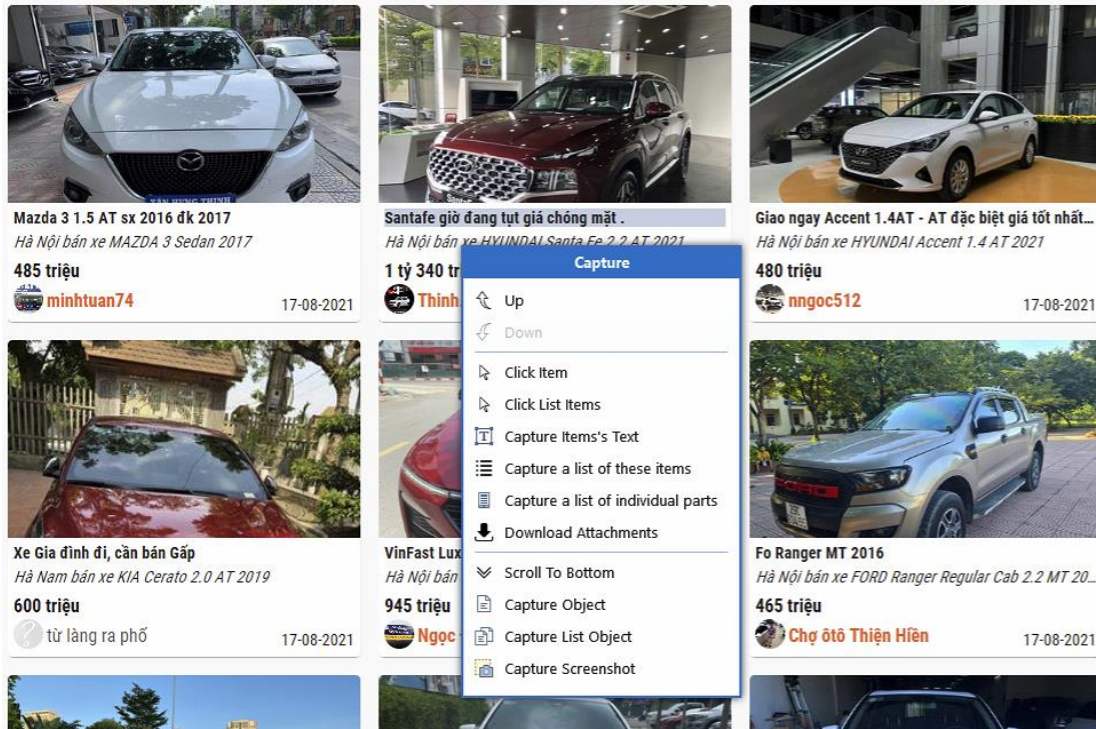


Figure 21 Workbench - Capture list of items

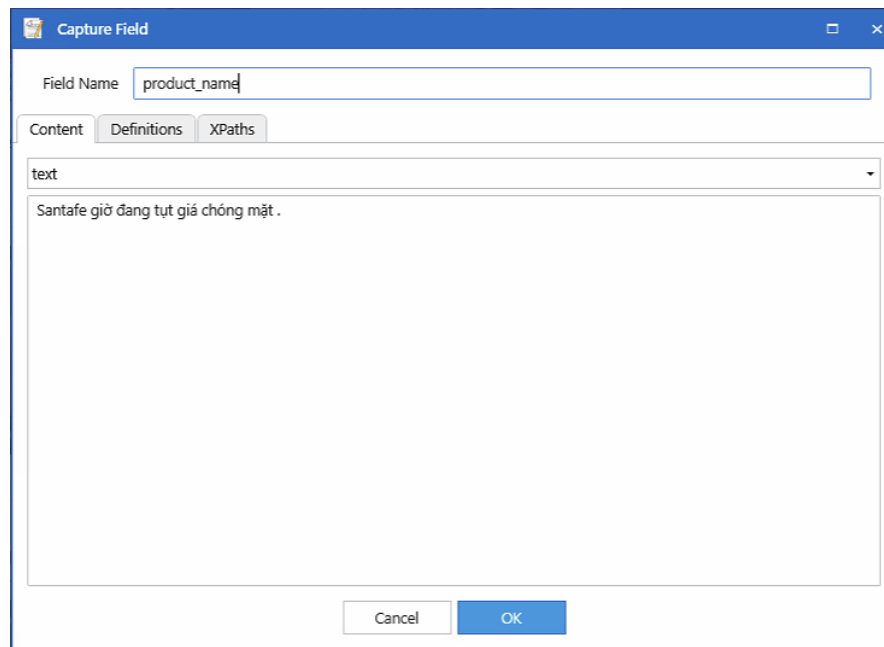


Figure 22 Workbench - Define data field

Chợ xe Otofun - Mua bán ô tô xe máy uy tín... Run test: Chợ xe Otofun - Mua bán ô tô xe máy uy tín...

Mercedes Benz C250, sản xuất 2013, đăng ký 2014. Biển Hà Nội bán xe MERCEDES BENZ C-Class 1.8AT AT 2013
660 triệu
Ngọc Đoàn auto 17-08-2021

Hyundai Solati 2021 - Solati - Cam kết Giá tốt nhất toàn hệ thống Hà Nội bán xe HYUNDAI Limosin 2.5 MT 2021
975 triệu
hyundai city 17-08-2021

Hyundai GRAND I10 2021 - GIẢM GIÁ SỐC - Cam kết Giá tốt nhất Hà Nội bán xe HYUNDAI I10 1.2 AT 2021
315 triệu
hyundai city 17-08-2021

Hyundai ACCENT model 2021 - Khuyến mãi lớn - CAM KẾT GIÁ TỐT NHẤT Hà Nội bán xe HYUNDAI Accent 1.4 AT 2021
400 triệu
hyundai city 17-08-2021

Hyundai KONA 2021 mẫu mới - GIÁ TỐT NHẤT - Hỗ trợ vay lãi...
Hyundai TUCSON model 2021 - Khuyến mãi lớn - GIÁ TỐT NHẤT...

Choxe OTOFUN Đăng ký Đăng nhập tìm kiếm...

Xe đang bán / HYUNDAI / Accent / Hà Nội

Hyundai ACCENT model 2021 - Khuyến mãi lớn - CAM KẾT GIÁ TỐT NHẤT

Hà Nội bán xe HYUNDAI Accent 1.4 AT 2021

Tình trạng: Mới
Giá bán: **400 triệu**
Đăng bởi hyundai city @ 21/11/19 10:01 ch **BẢO CẢO**

Thông số cơ bản

Hãng xe:	HYUNDAI	Mẫu xe:	Accent
Năm sản xuất:	2021	Số KM đã đi:	0 km (xe mới)
Màu sắc:	Màu khác	Hộp số:	Số tự động
Nhiên liệu:	Xăng	Động cơ:	1.4
Kiểu dẫn động:	Cầu trước (FWD)		

TỔNG CÔNG TY CÔNG NGHIỆP Ô TÔ VIỆT NAM - TẬP ĐOÀN Ô TÔ HYUNDAI THÀNH CÔNG

Hyundai Accent model 2021 mẫu mới ra mắt thị trường :

#	productname	productdescription	productprice	buyername	buyermobile
8	Mercedes E250 AMG, sản xuất 2015 Màu trắng	Hà Nội bán xe MERCEDES BENZ E-Class 1.8AT AT 2015	1 triệu 130 ngàn	ngọc đoàn auto @ Ngọc Đoàn auto	Mercedes E250 AMG, sản xuất 2015 Màu trắng ,nội thất
9	Mercedes Benz C250, sản xuất 2013, đăng ký 2014. Biển Hà Nội	Hà Nội bán xe MERCEDES BENZ C-Class 1.8AT AT 2013	660 triệu	ngọc đoàn auto @ Ngọc Đoàn auto	Mercedes Benz C250, sản xuất 2013, đăng ký 2014. Biển H
10	Hyundai Solati 2021 - Solati - Cam kết Giá tốt nhất toàn hệ thống	Hà Nội bán xe HYUNDAI Limosin 2.5 MT 2021	975 triệu	Hùng Hyundai @ hyundai city	TỔNG CÔNG TY CÔNG NGHIỆP Ô TÔ VIỆT NAM - TẬP Đ
11	Hyundai GRAND I10 2021 - GIẢM GIÁ SỐC - Cam kết Giá tốt nhất	Hà Nội bán xe HYUNDAI I10 1.2 AT 2021	315 triệu	Hyundai Hùng @ hyundai city	TỔNG CÔNG TY CÔNG NGHIỆP Ô TÔ VIỆT NAM - TẬP Đ
12	Hyundai ACCENT model 2021 - Khuyến mãi lớn - CAM KẾT GIÁ TỐT NHẤT	Hà Nội bán xe HYUNDAI Accent 1.4 AT 2021	400 triệu	Hyundai Hùng @ hyundai city	TỔNG CÔNG TY CÔNG NGHIỆP Ô TÔ VIỆT NAM - TẬP Đ

Figure 23: Workbench - Run simulation

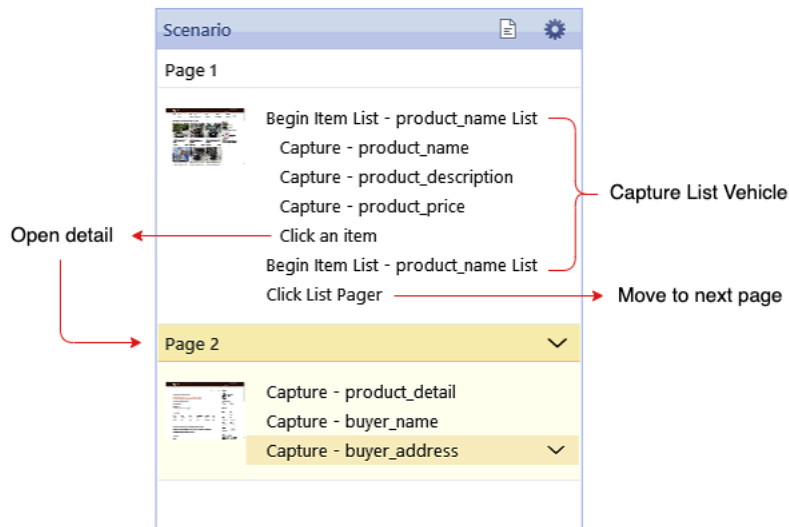


Figure 24 Workbench - Final Scenario

✓ Step 2: Save and execute

Figure 25 Workbench: Save project

✓ Step 3: Enjoy the results

Users can access the data after collection through desktop application or web application.

☐	agentid	product_name	product_description	product_price_text	product_price_val	product_detail	create_date	Action
☐	6354	Bán Toyota Camry 2.5Q màu đen mo...	Hà Nội bán xe TOYOTA Camry 2.5 AT ...	685 triệu	685000000	Toyota Camry 2.5Q model 2014 cực c...	2021-08-19 07:28:25	Delete
☐	6354	Bán Toyota Yaris G màu trắng sản xuấ...	Hà Nội bán xe TOYOTA Yaris 1.3 AT 2...	466 triệu	466000000	Chính chủ bán xe Yaris G 2014, đăng ...	2021-08-19 07:28:25	Delete
☐	6354	Bán Toyota Yaris 1.5AT màu trắng sản...	Hà Nội bán xe TOYOTA Yaris 1.5 AT 2...	369 triệu	369000000	Toyota Yaris 1.5AT sản xuất 2012, đã...	2021-08-19 07:28:24	Delete
☐	6354	Bán Kia Cerato 2.0AT màu trắng sản x...	Hà Nội bán xe KIA Cerato 2.0 AT 2016	490 triệu	490000000	Kia Cerato 2.0AT sản xuất 2016 bán f...	2021-08-19 07:28:21	Delete
☐	6354	Các dòng xe côn tay hót	Hà Nội bán xe YAMAHA R15 2019	66 triệu 666 ngàn	66666666	Những dòng xe hót trên hiện nay đã đ...	2021-08-19 07:28:09	Delete
☐	6354	Bán Toyota Innova 2.0V màu vàng đỏ...	Hà Nội bán xe TOYOTA Innova 2.0 AT ...	679 triệu	679000000	Toyota Innova bán đặc biệt 2.0V 2017...	2021-08-19 07:28:05	Delete
☐	6354	Bán LandCruiser Prado TXL 2.7 Mode...	Hà Nội bán xe TOYOTA Prado 2.7L AT...	830 triệu	830000000	- LandCruiser Prado TXL 2.7 Model 2...	2021-08-19 07:28:04	Delete
☐	6354	Bán Mercedes V250 Model 2017 Trăn...	Hà Nội bán xe MERCEDES BENZ V-Cl...	1 tỷ 650 triệu	1650000000	- Mercedes V250 Model 2017 Trắng n...	2021-08-19 07:28:04	Delete
☐	6354	Bán Acura MDX 3.7AWD , sản xuất 2...	Hà Nội bán xe ACURA MDX 3.7L AT 2...	440 triệu	440000000	Bán Acura MDX 3.7AWD , sản xuất 2...	2021-08-19 07:28:03	Delete
☐	6354	Bán Lexus GX460 sản xuất 2016	Hà Nội bán xe LEXUS GX 4.6L AT 2016	3 tỷ 450 triệu	3450000000	✓ Lexus GX460 sản xuất 2016Năng h...	2021-08-19 07:28:02	Delete
☐	6354	Bán xe Hyundai Starex độ Limousine ...	Hà Nội bán xe HYUNDAI Starex 2.5L ...	580 triệu	580000000	STAREX sx2014 độ limousine cực ...	2021-08-19 07:28:01	Delete
☐	6354	Bán BMW X6 E71 độ Bodykit Haman ...	Hà Nội bán xe BMW X6 3.0L AT 2008	700 triệu	700000000	- BMW X6 E71 bán độ bodykit Hama...	2021-08-19 07:27:59	Delete
☐	6354	Mercedes V250 up Maybach V cla...	Hà Nội bán xe MERCEDES BENZ V-Cl...	2 tỷ 450 triệu	2450000000	Mercedes V250 up Maybach V cla...	2021-08-19 07:27:57	Delete

Figure 26 Access collected data through web application

1.13 System Deployment & Result

1.13.1 System Deployment

I used 02 servers to install this system system. The detailed configuration is described in Table 4

#	CPU	Core	RAM	HDD	OS
1	Intel 2x CPU E5-2680v3	24	32G RAM	8T	CentOS Linux 7 (Core)
2	Intel 2x CPU E5-2680v3	24	32G RAM	8T	CentOS Linux 7 (Core)

Table 4 Servers configuration

Các phần mềm và dịch vụ trên mỗi server được mô tả trong bảng Table 5

#	Software/Service	Version
Server #1		
1	OpenJDK	1.8.0_252
2	MongoDB	4.2.8
3	PostgreSQL	9.2.24
Server #2		
1	OpenJDK	1.8.0_252
2	Extractor-Manager	
3	Extractor-System-Monitor	
4	Extractor-Worker	
5	Extractor-Web	

Table 5 System settings information

1.13.2 Results

This system is experimentally run on the data source of popular online newspapers and forums in Vietnam, specifically including **114** online newspapers and **18** forums, with a total of **4,693** category links. The update frequency for each category link is from **2 ~ 3 hours**. The delay between the actual time a job is run and the expected scheduled time is about **5 - 10 seconds**. On average, there are about **25,000 ~ 30,000** new posts per day. Total number of articles on both channels is **32,812,697**.

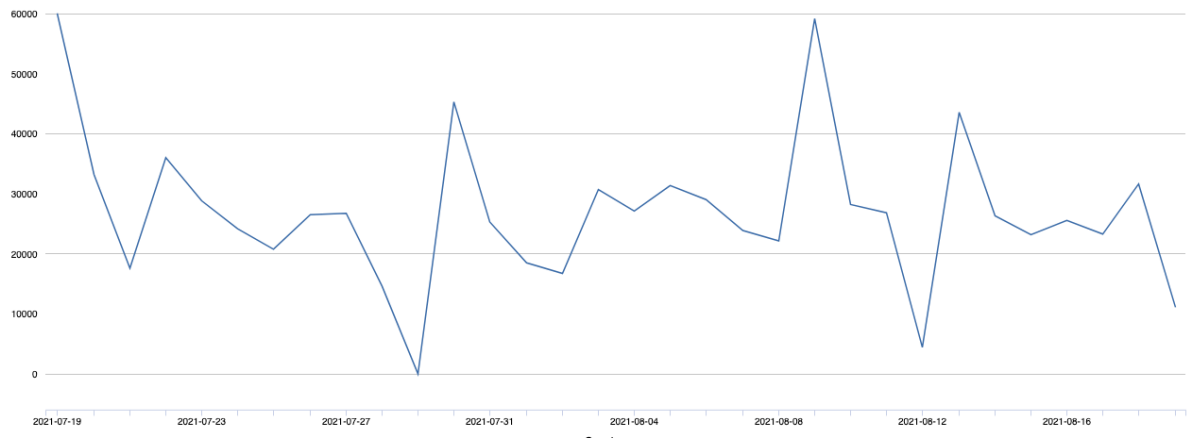


Figure 27 Statistics of new posts by day

☐	AgentId	Name	URL	Category	Schedule	Status	Datas
☐	16	ALL	https://dantri.com.vn/			new	244,784
☐	17	Su kien	https://dantri.com.vn/su-kien.htm		Interval: 2 hours	completed	10,721
☐	164	Xa hoi	https://dantri.com.vn/xa-hoi.htm		Interval: 2 hours	stopped	7,524
☐	166	The gioi	https://dantri.com.vn/the-gioi.htm		Interval: 2 hours	completed	5,093
☐	167	The theo	https://dantri.com.vn/the-thao.htm		Interval: 2 hours	completed	6,129
☐	168	Anh sinh	https://dantri.com.vn/an-sinh.htm		Interval: 2 hours	completed	3,632
☐	169	Suc khoe	https://dantri.com.vn/suc-khoe.htm		Interval: 2 hours	stopped	5,177
☐	170	Nhan ai	https://dantri.com.vn/tam-long-nhan-ai.htm		Interval: 2 hours	stopped	2,966
☐	171	Kinh doanh	https://dantri.com.vn/kinh-doanh.htm		Interval: 2 hours	completed	4,934

Figure 28 Number of articles by category of dantri newspaper

Chapter 4

Social Listening for Crisis Management

With the advent of the social web, users express their opinions and comments on social media. This user-generated information has become a valuable asset to understand people's opinions and can leverage the insights obtained from surveys with a targeted population. Nevertheless, to understand the wisdom of the crowd, it needs to develop technologies to collect, filter, analyze and visualize social media. Social media monitoring tools provide these facilities.

This chapter presents a social network monitoring tool, as a concrete demonstration of the practicality of the data collection system presented in chapter 3.

1.14 Requirement Analysis

Nowadays, the Internet has become a huge data warehouse, where a consumer can freely share their opinions and feelings about various topics with large crowds; it is also where a businessman can collect and analyze consumer tastes thereby helping them come up with good business strategies. Data on the internet, especially from news, forums and documents on social networks is being widely used for various purposes:

Consumer behavior analytics help the businessman can fully understand customer needs and behaviors in order to develop a single view of the customer and thereby build compelling, differentiated offers throughout the sale and ownership cycle which are relevant in today's digital environment [22], [23]. Sentiment Analysis is considered one of the most attractive fields that encourage to study and apply in various sectors. Analyzing consumers' opinions using Social Networking Service (SNS) media data can be a very great way for automotive companies to enhance their marketing targets and objectives [24], [25].

Marketing analytics, trend analytics has the potential to significantly improve the decision making of businessman and returns they can deliver, by collating and analyzing marketing information and customer behaviors in a consistent and seamless environment. The papers [26], [27] providing insights to assist in answering critical questions such as: what is the most suitable product for a specific market; how to advertise such product in that market;

through what communication channels; at what points in time and for what price; and supported by what kind of promotional and advertising actions.

Discovering Hot Topic in online e-commerce communities help producer, sales, marketer better understand customers' need and interest in product-related knowledge. The paper [28] have proposed the Social Listening System, which enables anyone to access useful information. In addition to the keyword-based features used in traditional text clustering, the authors integrate auto domain-specific features to represent the messages posted in online health communities. This system help discovers hot topics in SNS from various perspective depending users point of view such as a hint of product planning, design of the questionnaire, risk management, etc.

Crisis management in cyberspace is becoming increasingly important as it directly affects the survival of the business. When a crisis occurs, if businesses do not handle it wisely, the damage will be huge. However, if it is possible to identify and promptly warn potential problems, businesses can promptly prevent unnecessary incidents. With this in mind, there are a variety of ways that social listening can support the crisis management process [29], [30], [31], [32].

This chapter will present the social listening system applied in fighting communication crisis. The main goal of the system is to track publicly published articles on online news sites, forums, e-commerce sites and personal blogs. Then, through the analysis of these data (emotion analysis, prominent topic analysis), the system will point out potential problems (negative posts, topics of interest) to help. businesses make plans and handling strategies quickly and in a timely manner.

1.15 Architecture

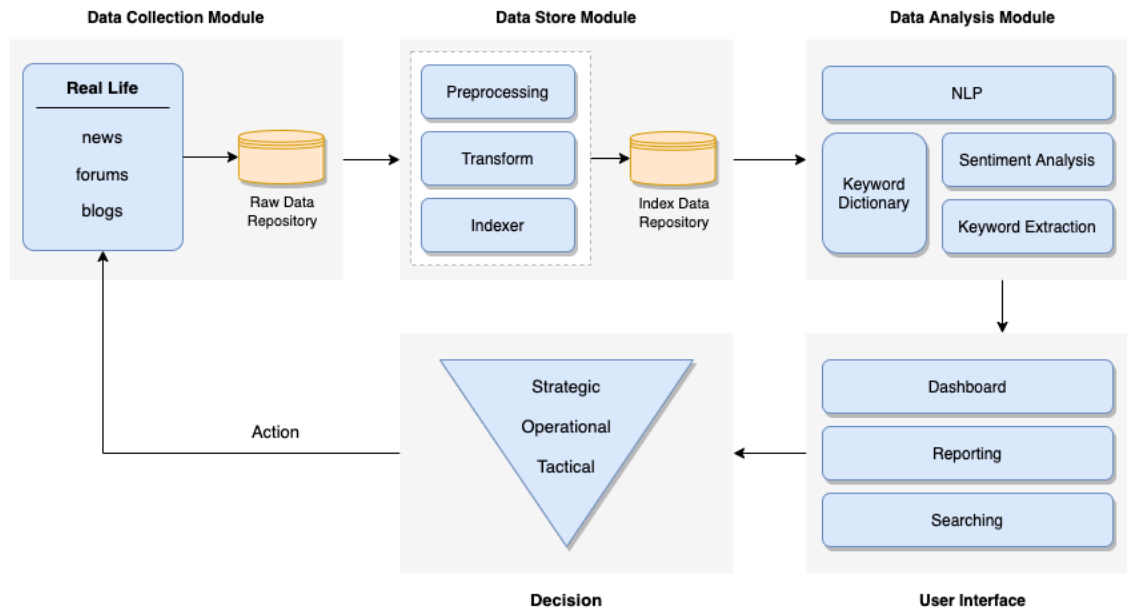


Figure 29 The social listening architecture

The architecture of the social listening for monitoring crisis comprising four individual modules (Figure 19)

- (1) **The Data collection module** allows automatic or semi-automatic collection of social media data for the purpose of digital data collections. From a large data perspective, data collection is difficult to achieve. Not just the huge volume of data is a problem, but their format also generates difficulties in the collection process. This module has been detailed in Chapter 3.
- (2) **The Data storage module** have the role of cleaning and converting raw data collected from many sources into a normalized form. The data is then indexed and stored for later retrieval and computation.
- (3) **The data analysis module** includes components of natural language processing, nuance analysis, statistics, etc., which is considered as the brain of the whole system.
- (4) **User Interface:** The application enables users to discover how media is shared in Online Social Networks. A user can create several campaigns, each one linked to his/her

preferences. To initiate a collection, the user must define a set of keywords, all relevant to a topic user wants to monitor.

1.16 Enabling Technologies

1.16.1 Elasticsearch

Due to the rapid development of network technology, the amount of data generated in the Internet every day is very large, and we want to collect, store and efficiently retrieve the data of media resources in the network.

Elasticsearch is a distributed, open-source search and analytics engine built on Apache Lucene and developed in Java. It started as a scalable version of the Lucene open-source search framework then added the ability to horizontally scale Lucene indices. Elasticsearch allows you to store, search, and analyze huge volumes of data quickly and in near real-time and give back answers in milliseconds. It's able to achieve fast search responses because instead of searching the text directly, it searches an index

1.16.2 Natural Language Processing (NLP)

Natural language processing (NLP) is the ability of a computer program to understand human language as it is spoken and written -- referred to as natural language. It is a component of artificial intelligence (AI). NLP enables computers to understand natural language as humans do. Whether the language is spoken or written, natural language processing uses artificial intelligence to take real-world input, process it, and make sense of it in a way a computer can understand.

1.16.3 Sentiment Analysis

Sentiment Analysis is the process of determining whether a piece of writing is positive, negative or neutral. A sentiment analysis system for text analysis combines natural language processing (NLP) and machine learning techniques to assign weighted sentiment scores to the entities, topics, themes and categories within a sentence or phrase.

Crisis Management is an important area of business for Public Relations Specialist (PRS) and sentiment analysis is one of their tool for everyday use. Using sentiment analysis, PRS can receive real time notifications about any negative piece of content that appeared

online. On seeing a negative customer sentiment mention, a company can quickly react and nip the problem in the bud before it escalates into a social media crisis.

There are lots of different ways to perform sentiment analysis, and using a dictionary is possibly the simplest one. This is the way I used it in this thesis.

Dictionary-based sentiment analysis is a computational approach to measuring the feeling that a text conveys to the reader. This method relies heavily on a pre-defined list (or dictionary) of sentiment-laden words. A dictionary usually provides one or more scores for each word. They are then used to calculate the overall feeling of a sentence based on single words.

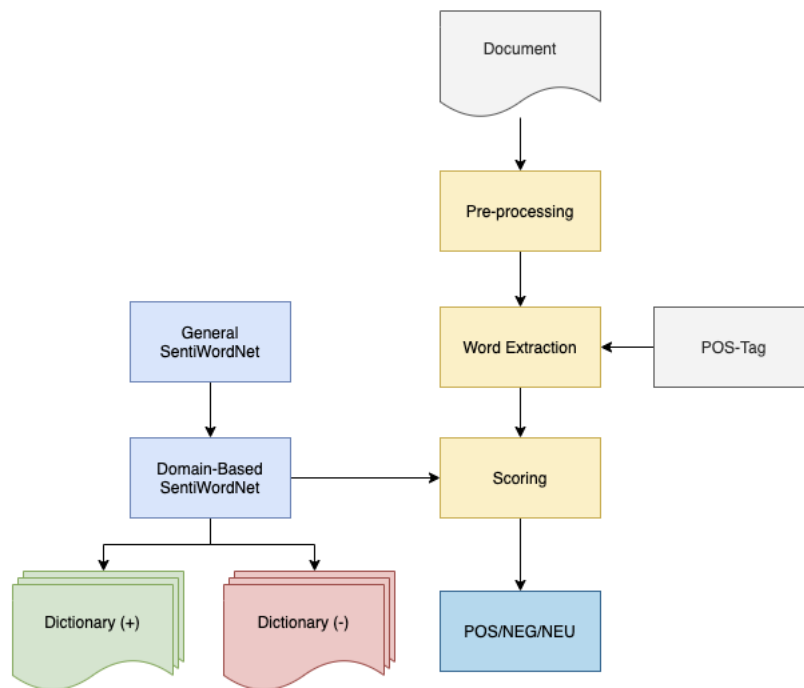


Figure 30 Dictionary-base sentiment analysis flow

1.16.4 Keyword Extraction

Keyword Extraction Overview is a text analysis technique. We can obtain important insights into the topic within a short span of time. It helps concise the text and obtain relevant keywords. It saves the time of going through the entire document. Example use-cases are finding topics of interest from a news article and identifying the problems based on customer reviews and so. One of the techniques used for Keyword Extraction is TextRank.

TextRank is a graph based algorithm for Natural Language Processing that can be used for keyword and sentence extraction. The algorithm is inspired by PageRank which was used by Google to rank websites.

PageRank (PR) is an algorithm used to calculate the weight for web pages. We can take all web pages as a big directed graph. In this graph, a node is a webpage. If webpage A has the link to web page B, it can be represented as a directed edge from A to B.

For a webpage V_i , $In(V_i)$ is a set of webpages point to it while V_j is the set of vertices V_i point to. The rank of V_i is defined as:

$$S(V_i) = (1 - d) + d * \sum_{j \in In(V_i)} \frac{S(V_j)}{|Out(V_j)|}$$

- $S(V_i)$ – the weight of webpage i
- d – damping factor (between 0 and 1), in case of no outgoing links
- $In(V_i)$ – inbound link of i , which is a set
- $Out(V_j)$ – outgoing link of j , which is a set
- $|Out(V_j)|$ - the number of outbound links

In case of text, the connections can be weighted. We introduced w_{ij} which is the weight of the edge between entity V_i and V_j . The weighed rank of V_i now becomes:

$$WS(V_i) = (1 - d) + d * \sum_{V_j \in In(V_i)} \frac{w_{ji}}{\sum_{V_k \in Out(V_j)} w_{jk}} WS(V_j)$$

For keyword extraction we want to identify a subset of terms that best describe the text. We follow these steps:

1. Tokenize and annotate with Part of Speech (PoS). Only consider single words. No n-grams used, multi-words are reconstructed later.
2. Use syntactic filter on all the lexical units (e.g. all words, nouns and verbs only).
3. Create an edge if lexical units co-occur within a window of N words to obtain an unweighted undirected graph.

4. Run the text rank algorithm to rank the words.
5. We take the top lexical words.
6. Adjacent keywords are collapsed into a multi-word keyword.

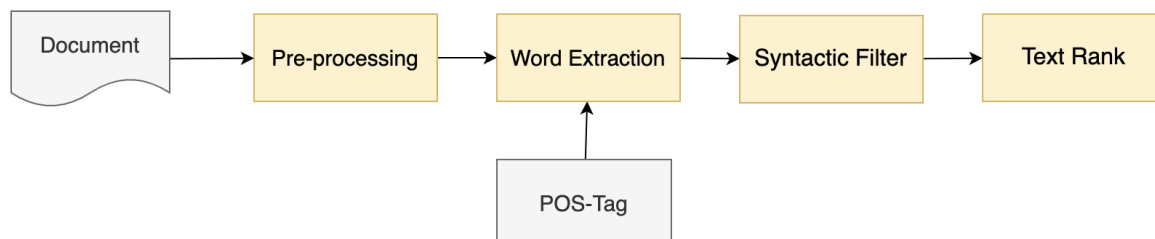


Figure 31 Keywords extraction flow

1.17 Case Study

Consider the recent case of BKAV being attacked by hackers. The data were analyzed for a period of one month from July 18, 2021 to August 19, 2021.

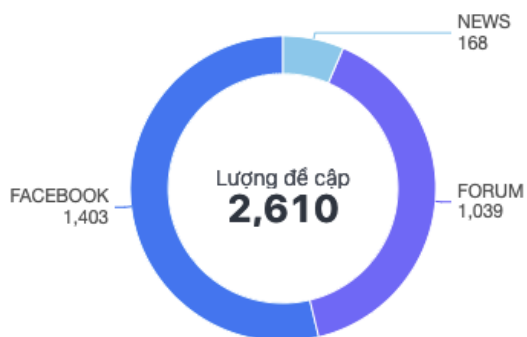


Figure 32 Mentions of BKAV by channel

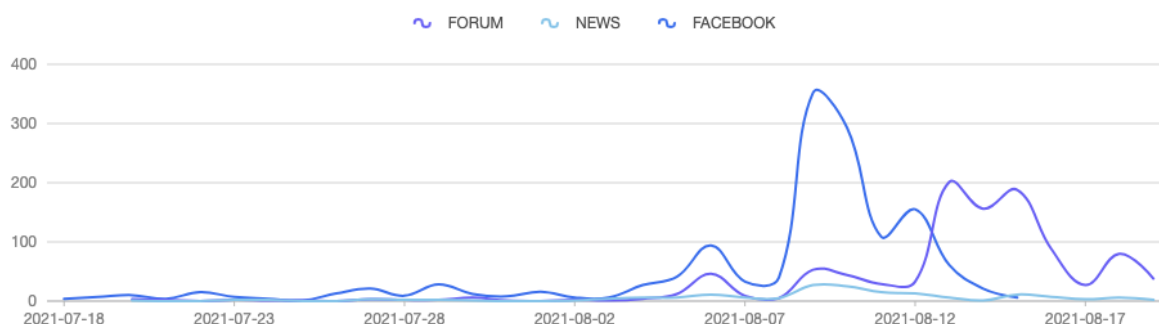


Figure 33 Mentions of BKAV over time

Chapter 5

Conclusion and Future Work

The thesis has detailed the process of building a data collection system according to the scenario on the web. From the evaluation and analysis obtained through the process of researching and surveying existing data collection systems, the thesis proposes a solution to build a new system that fully meets the requirements that existing systems have. system for collecting and storing large amounts of data. The newly built system has outstanding advantages over existing systems such as the ability to run on multiple operating system environments, the ability to run multi-threading, the ability to share work, the ability to scale the system on multiple machines.

At the same time, the social network listening system is also presented as a demonstration of the effectiveness and practicality of the data collection system. That is just one use case, in addition, the data collection system can also serve many different needs such as: Used car pricing system based on the selling price on the online car market; Data Enrichment for Machine Learning Systems...

In the process of running the actual test, the system also arises a few unoptimized problems, which need to be researched and improved in the near future.

- The environment is not homogenous between the process of running the emulator when creating the script and the real environment on the server, leading to many situations where the emulator runs well but cannot run on the server.
- Xpath generation algorithm in many cases is not good.
- System installation is still complicated.
- The data quality monitoring system is still rudimentary and has to be done manually.

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